BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 1

FUNDAMENTALS OF MATHEMATICS

Unit 6

Special Functions Part 1

Table of Contents

SL No	Topic		Fig No / Table / Graph	SAQ / Activity	Page No
1	Introduction		-	-	3-4
	1.1	Learning Objectives	-	-	
2	Radian or Circular Measure		-	-	5-6
3	Trigonometric Functions		-	-	7-10
	3.1	Graphical Representation of Trigonometric Functions	-	-	
4	Trigonometrical ratios of certain standard angles		-	-	11
5	Allied Angles and Compound Angles		-	-	12-13
	5.1	Allied Angles	-	-	
	5.2	Compound Angles	-	-	
6	Inverse Trigonometric Functions		-	-	14-15
7	Exponential Functions		-	-	16-17
	7.1	Properties of Exponential functions	-	-	
	7.2	Derivatives of Exponential Functions	-	-	
8	Summary		-	-	18-19
9	Self-Assessment Questions		-	1	19-21
10	Terminal Questions		-	-	22
11	Answers		-	-	23-24
12	References		-	-	24

1. INTRODUCTION

In Unit 5, we embarked on a comprehensive exploration of Continuity and Differentiability, dissecting the nuanced interplay between these fundamental concepts and their implications on the behaviour of functions. Our journey delved into the intricacies of composite functions and unravelled the chain rule, a pivotal tool in calculus for dissecting the derivatives of these complex entities. We ventured into the realm of inverse trigonometric functions and implicit functions, broadening our understanding of derivatives beyond conventional boundaries. The unit also illuminated the profound concepts of exponential and logarithmic functions, enriching our comprehension of their derivatives through the lens of logarithmic differentiation. The study extended to the derivative of functions expressed in parametric forms and the profound insights offered by second-order derivatives. Culminating with Rolle's and Lagrange's Mean Value Theorems, the unit not only provided a robust mathematical foundation but also painted a vivid picture of these theorems' geometric interpretations, enriching our conceptual and visual understanding of calculus.

As we transition into Unit 6: Some Special Functions Part 1, we are set to navigate the realm of Radian or Circular Measure and Trigonometric Functions. This unit is instrumental for computer application students, as it lays the foundational stones for understanding the mathematical underpinnings of various algorithms, particularly those involving geometric calculations, rotations, and transformations integral to computer graphics, simulations, and data analysis. The precision and clarity offered by radian measures in articulating angular displacements and rotations are indispensable for programming and developing applications that are both mathematically accurate and visually coherent.

To effectively engage with this unit, a strategic approach to learning is paramount. It begins with a solid grasp of the definitions and properties of radian measures and trigonometric functions, ensuring a robust conceptual framework. Visual representations and graphical interpretations play a crucial role in internalizing the relationships and transformations described by these functions. Practicing the application of these concepts through real-world problems and programming challenges will not only reinforce understanding but also highlight their practical utility in computer science. By intertwining theoretical knowledge with practical applications, this unit promises to augment our mathematical arsenal,

enhancing our ability to solve complex problems and develop innovative solutions in the realm of computer applications.

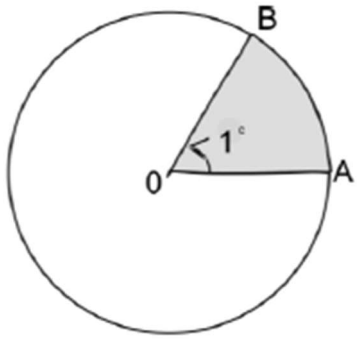
1.1 Learning Objectives

At the end of this unit, you will be able to:

- *Define the concept of a radian and describe its relationship to degrees.*
- *Explain the significance of the unit circle in defining trigonometric functions.*
- *Calculate the trigonometric ratios for standard angles using the unit circle.*
- *Differentiate between the trigonometric functions and their reciprocal functions in terms of their relationships and applications.*

2. RADIAN OR CIRCULAR MEASURE

A radian is the angle subtended at the centre of a circle by an arc equal to the radius of the circle. O is the centre of a circle. A and B are points on the circle such that arc AB = radius OA. Then ∠AOB is called one radian or one circular measure. We write ∠AOB = 1°.



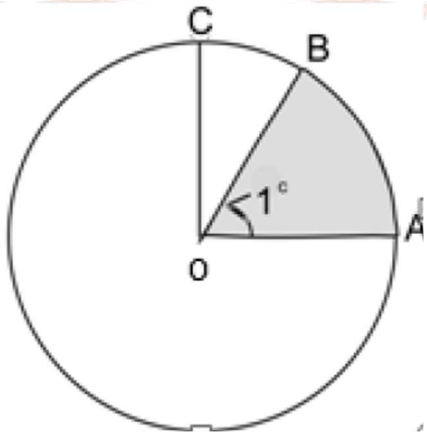
Angle AOB

Radian is a constant angle and $\pi^c = 180^0$

Consider a circle whose centre is O and radius r. A and B are points on the circle such that arc AB = OA = r. Join OA, OB and draw OC perpendicular to OA. ∠AOB = 1^c and ∠AOC= 1 right angle. We know that arc $AC = \frac{1}{4}$(circumference of the circle) $= \frac{1}{4}(2\pi r) = \frac{\pi r}{2}$. In a circle the arcs are proportionated to the angles subtended by them at

Circle the centre.

i.e.



$$\frac{arcAB}{arcAC} = \frac{\angle AOB}{\angle AOC}$$

$$\frac{r}{\pi r\ /\ 2} = \frac{1^c}{1\ right\ angle}$$

$$\frac{2}{\pi} = \frac{1^c}{1\ right\ angle}$$

Therefore,

$$1^c = \frac{2}{\pi \cdot 1\ right\ angle}$$

which is constant.

Therefore, we can conclude that Radian is a constant angle.

Furthermore, we have

$$\pi \cdot 1^c = 2 \cdot 1\ right\ angle$$

$$\pi^c = 2 \times 90^0 = 180^0$$

Note:

1. $\pi^c = 180^0$ means π radians are equal to 180^0. Hereafter, this will be written as $\pi = 180^0$.

   For example $\frac{\pi}{2} = 90^0, \frac{\pi}{4} = 45^0, 2\pi = 360^0$ and so on

2. $1^c = \frac{180}{\pi} = \frac{180}{217} = \frac{90 \cdot 7}{11} = \frac{630}{11} = 57^0 17' 45''$ (nearly)

   Therefore, we can conclude that 1 radian is < than 60°

3. TRIGONOMETRIC FUNCTIONS

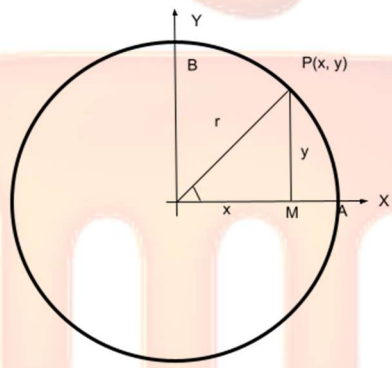
Consider a circle whose centre is the origin and radius is r. Let the circle cut X-axis at A and Y-axis at B and P(x, y) is any point on the circle. Join OP and draw PM ⊥ to X-axis.

OP = r

OM = x

MP = y

∠XOP = θ



Unit Circle

The six trigonometrical functions (ratios) of angle Θ are defined as:

Sine of angle Θ = $sin\,\theta = \frac{y}{r}$

Cosine of angle Θ = $cos\,\theta = \frac{x}{r}$

Tangent of angle Θ = $tan\,\theta = \frac{y}{x}(x \neq 0)$

Cosecant of angle Θ = $cosec\,\theta = \frac{r}{y}(y \neq 0)$

Secant of angle Θ = $sec\,\theta = \frac{r}{x}(x \neq 0)$

Cotangent of angle Θ = $cot\,\theta = \frac{x}{y}(y \neq 0)$

Since,

$$r^2 = x^2 + y^2$$

we have

$$r = \pm\sqrt{x^2 + y^2}$$

So, we have

$$sin\,\theta = \frac{y}{\sqrt{x^2+y}}2$$

$$cos\,\theta = \frac{x}{\sqrt{x^2+y^2}}$$

$$tan\,\theta = \frac{y}{x}(x \neq 0)$$

$$cot\,\theta = \frac{x}{y}(y \neq 0)$$

$$sec\,\theta = \frac{\sqrt{x^2+y^2}}{x}(x \neq 0)$$

$$cosec\,\theta = \frac{\sqrt{x^2+y^2}}{y}(y \neq 0)$$

Points to remember

1. Reciprocal relations:

$$sin\,\theta : cos\,ec\theta = \frac{y}{r} : \frac{r}{y} = 1$$

Therefore, sinΘ and cosecΘ are reciprocal to each other.

i.e.:

$$sin\,\theta = \frac{1}{cosec\,\theta}$$

and

$$cosec\,\theta = \frac{1}{sin\,\theta}$$

Similarly,

$$cos\,\theta = \frac{1}{sec\,\theta}$$

$$sec\,\theta = \frac{1}{cos\,\theta}$$

$$tan\,\theta = \frac{1}{cot\,\theta}$$

$$cot\,\theta = \frac{1}{tan\,\theta}$$

2.

$$\frac{sin\,\theta}{cos\,\theta} = \frac{y/r}{x/r} = \frac{y}{x} = tan\,\theta$$

Therefore,

$$tan\,\theta = \frac{sin\,\theta}{cos\,\theta}$$

$$cot\,\theta = \frac{cos\,\theta}{sin\,\theta}$$

3. The above definitions of trigonometric functions hold good whatever may be the position of the point P(x, y) on the circle.
4. Identities
   a. $sin^2\,\theta + cos^2\,\theta = 1$
   b. $1 + tan^2\,\theta = sec^2\,\theta$
   c. $1 + cot^2\,\theta = cosec^2\,\theta$
   d. From the above figure:
      i.

$$x^2 + y^2 = r^2$$

$$on\ dividing\ by\ r^2$$

$$\frac{x^2}{r^2} + \frac{y^2}{r^2} = 1$$

But,

$$\frac{x}{\gamma} = cos\,\theta$$

and

$$\frac{y}{\gamma} = sin\,\theta$$

Therefore,

$$cos^2\,\theta + sin^2\,\theta = 1$$

ii.

$$x^2 + y^2 = r^2$$

If $x \neq 0$, we can divide by x^2

$$1 + \frac{y^2}{x^2} = \frac{r^2}{x^2}$$

But,

$$\frac{x}{y} = tan\,\theta$$

And

$$\frac{r}{x} = sec\,\theta$$

Therefore,

$$1 + tan^2\,\theta = sec^2\,\theta$$

iii.

$$x^2 + y^2 = x^2$$

If $y \neq 0$, we can divide by y^2

$$\frac{x^2}{y^2} + 1 = \frac{r^2}{y^2}$$

But,

$$\frac{x}{y} = cot\,\theta$$

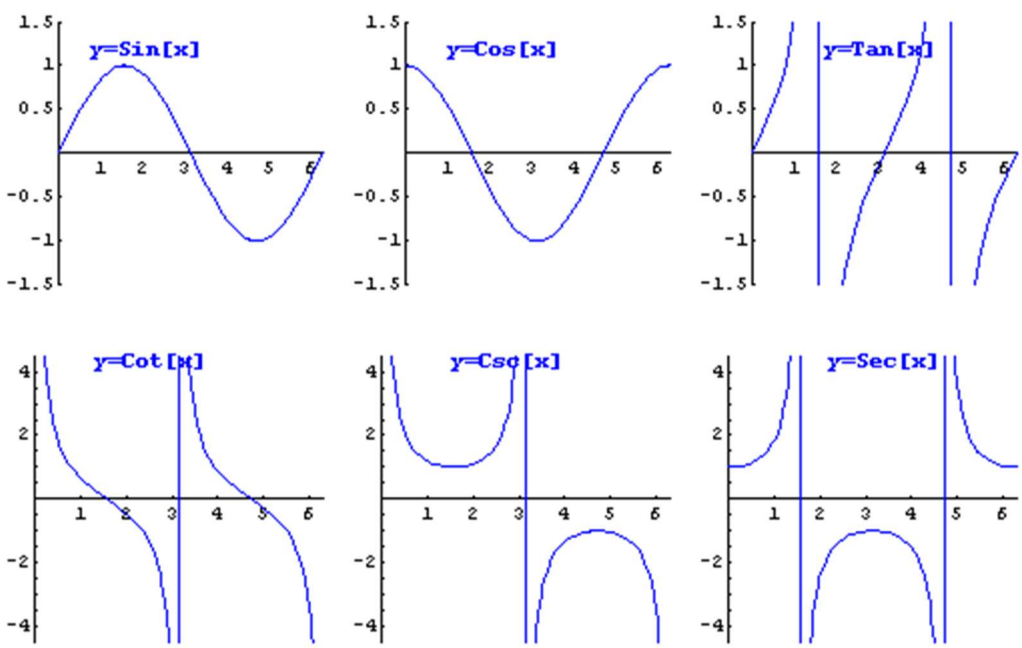
and

$$\frac{r}{y} = cos\,ec\theta$$

Therefore,

$$cot^2\,\theta + 1 = cosec^2\,\theta$$

Graphical Representation of Trigonometric Functions



Graphical Representation of Trigonometric Functions

4. TRIGONOMETRICAL RATIOS OF CERTAIN STANDARD ANGLES

To remember the trigonometrical ratios of the standard angles the following table is useful.

	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞

Example:

Find the value of

$$\frac{tan^2\,60 - 2\,tan^2\,45 + sec^2\,30}{2\,sin^2\,45^0\,sin\,90 + cos^2\,60\,cos^2\,30}$$

(All angles are in degrees)

Solution:

$$\frac{tan^2\,60 - 2\,tan^2\,45 + sec^2\,30}{2\,sin^2\,45^0\,sin\,90 + cos^2\,60\,cos^2\,30} = \frac{\left(\sqrt{3}\right)^2 - 2(1)^2 + \left(2\,/\,\sqrt{3}\right)^2}{2\left({}^{1}/_{\sqrt{2}}\right)^2 \cdot 1 + \left({}^{1}/_{2}\right)^2 \cdot \left({}^{\sqrt{3}}/_{2}\right)^2}$$

$$\frac{(3-2) + (4\,/\,3)}{1 + (3\,/\,16)} = \frac{7\,/\,3}{19\,/\,16} = \frac{112}{57}$$

5. ALLIED ANGLES AND COMPOUND ANGLES

[All angles are in degrees]

Allied Angles

Connecting the trigonometrical ratios of -Θ with those of Θ

$$sin(-\theta) = sin\,\theta$$
$$cos(-\theta) = cos\,\theta$$
$$tan(-\theta) = -\,tan\,\theta$$
$$cosec(-\theta) = -\,cosec\,\theta$$
$$sec(-\theta) = sec\,\theta$$
$$cot(-\theta) = -\,cot\,\theta$$

Connecting the trigonometrical ratios of 90-Θ with those of Θ

$$sin(90-\theta) = cos\,\theta$$
$$cos(90-\theta) = sin\,\theta$$
$$tan(90-\theta) = cot\,\theta$$
$$cosec(90-\theta) = sec\,\theta$$
$$sec(90-\theta) = cosec\,\theta$$
$$cot(90-\theta) = tan\,\theta$$

Connecting the trigonometrical ratios of 90+Θ with those of Θ

$$sin(90+\theta) = cos\,\theta$$
$$cos(90+\theta) = -\,sin\,\theta$$
$$tan(90+\theta) = -\,cot\,\theta$$
$$cosec(90+\theta) = sec\,\theta$$
$$sec(90+\theta) = -\,cosec\,\theta$$
$$cot(90+\theta) = -\,tan\,\theta$$

Connecting the trigonometrical ratios of 180-Θ with those of Θ

$$sin(180-\theta) = sin\,\theta$$
$$cos(180-\theta) = -\,cos\,\theta$$
$$tan(180-\theta) = -\,tan\,\theta$$
$$cosec(18\theta-\theta) = cosec\,\theta$$

$$sec(180-\theta) = -sec\,\theta$$

$$cot(180-\theta) = -cot\,\theta$$

Connecting the trigonometrical ratios of 180+Θ with those of Θ

$$sin(180+\theta) = -\sin\theta$$

$$cos(180+\theta) = -cos\,\theta$$

$$tan(180+\theta) = tan\,\theta$$

$$cosec(18\theta+\theta) = -\operatorname{cosec}\theta$$

$$sec(180+\theta) = -sec\,\theta$$

$$cot(180+\theta) = cot\,\theta$$

Example:

Find

$$\frac{sin\,135 + cos\,480}{sin\,135 - cos\,120}$$

Solution:

$$sin\,135 = sin(180-45) = sin\,45 = \frac{1}{\sqrt{2}}$$

$$cos\,480 = cos(360+120) = cos\,120 = \cos(90+30) = -\sin 30 = -\frac{1}{2}$$

$$\frac{sin\,135 + cos\,480}{sin\,135 - cos\,120} = \frac{\frac{1}{\sqrt{2}} - \frac{1}{2}}{\frac{1}{\sqrt{2}} + \frac{1}{2}} = \frac{2-\sqrt{2}}{2+\sqrt{2}} = 3 - 2\sqrt{2}$$

Compound Angles

The sum of difference of angles like A+B, A-B, A+B-C, etc., are known as compound angles. The trigonometrical ratios of A+B, A-B can be expressed in terms of A and B.

1. $sin(A+B) = sin\,A\,cos\,B + cos\,A\,sin\,B$
2. $cos(A+B) = cos\,A\,cos\,B - sin\,A\,sin\,B$
3. $tan(A+B) = \frac{tan\,A + tan\,B}{1 - tan\,A\,tan\,B}$
4. $\sin(A-B) = \sin A \cos B - \cos A \sin B$
5. $cos(A-B) = cos\,A\,cos\,B + sin\,A\,sin\,B$
6. $tan(A-B) = \frac{tan\,A - tan\,B}{1 + tan\,A\,tan\,B}$

6. INVERSE TRIGONOMETRIC FUNCTIONS

Inverse trigonometric functions, also known as arc functions, are crucial in extending the utility of trigonometry beyond the realm of right-angled triangles. These functions serve as the inverse operations of the basic trigonometric functions, allowing us to find angle measures given specific trigonometric ratios. In this section we will look at the definitions, ranges, domains, and principal value branches of inverse trigonometric functions, alongside their graphical representations and elementary properties.

Definitions and Notations

Inverse trigonometric functions are denoted as follows:

- Inverse Sine: $sin^{-1}(x)$ or $asin(x)$
- Inverse Cosine: $cos^{-1}(x)$ or $acos(x)$
- Inverse Tangent: $tan^{-1}(x)$ or $atan(x)$
- Inverse Cotangent: $cot^{-1}(x)$ or $acot(x)$
- Inverse Secant: $sec^{-1}(x)$ or $asec(x)$
- Inverse Cosecant: $cosec^{-1}(x)$ or $acos\,ec(x)$

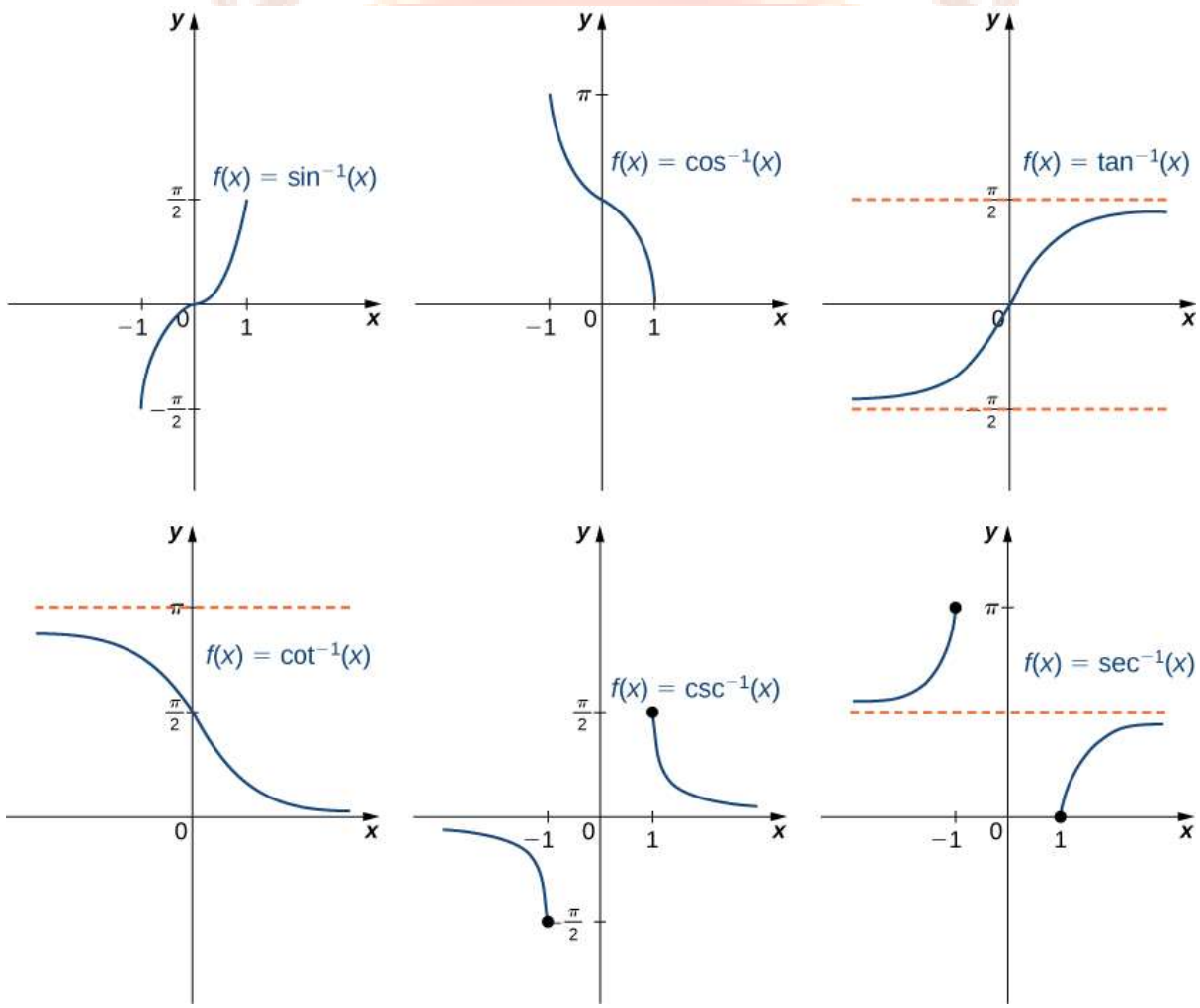
To ensure that each value of an inverse trigonometric function corresponds to exactly one angle, the outputs of these functions are restricted to principal values.

- $sin^{-1}(x)$: Domain [-1, 1], Range $[-\frac{\pi}{2}, \frac{\pi}{2}]$
- $cos^{-1}(x)$: Domain [-1, 1], Range $[0, \pi]$
- $tan^{-1}(x)$: Domain $\mathbb{R}(all\ real\ numbers)$. Range $[-\frac{\pi}{2}, \frac{\pi}{2}]$

Inverse trigonometric functions bridge the gap between angles and ratios, allowing for the determination of unknown angles in various contexts. Their definitions within the confines of the unit circle, coupled with specific domain and range restrictions, ensure that they provide unique and meaningful results. Understanding their graphs and elementary properties enhances one's ability to apply these functions effectively.

These functions not only serve as foundational mathematical tools but also find extensive application in fields requiring geometric computations, such as physics, engineering, and computer graphics. The examples provided illustrate the practicality of these functions in solving equations and finding angles, emphasizing their significance in both theoretical and applied mathematics. Through the study of inverse trigonometric functions, one gains a deeper appreciation for the interconnectivity of mathematical concepts and their applicability to real-world problems.

Graphical representation of Inverse Trigonometric Functions



7. EXPONENTIAL FUNCTIONS

Exponential functions represent a fundamental concept in mathematics, characterized by their unique property where the rate of change is proportional to the function's value at every point. This distinctive feature renders them invaluable across various domains, including natural sciences, economics, and engineering, particularly in modelling growth processes and decay.

An exponential function is defined as a function of the form $f(x) = a^x$, where:

- a is the base of the exponential function and is a positive constant
- x is the exponent, which can be any real number

The base a is crucial in determining the function's behavior. The most commonly used base is the natural number e (approximately equal to 2.71828), known for its unique properties in calculus.

Properties of Exponential functions

Exponential functions exhibit several key properties that make them distinct and widely applicable:

1. Positive Range: For $a > 0, a^x > 0$ for all x. This means the exponential function never touches or crosses the x-axis, implying it always yields positive values.
2. Continuity and Differentiability: Exponential functions are continuous and differentiable for all real numbers, which means they have no breaks, jumps, or sharp corners, and you can find their slope at any point.
3. Growth and Decay: If $a > 1$, the function represents exponential growth as it increases rapidly as x increases. Conversely, if $0 < a < 1$, the function depicts exponential decay, decreasing rapidly as x increases.
4. Asymptotic to the X-Axis: As x approaches negative infinity, a^x approaches 0, meaning the graph of an exponential function gets closer to the x-axis but never actually touches it.
5. Multiplicative Property: $a^{x+y} = a^x \cdot a^y$. This property is essential in simplifying expressions involving exponential functions.

6. Invertibility: Exponential functions are invertible, with their inverses being logarithmic functions. This interplay between exponential and logarithmic functions is fundamental in solving equations involving exponentials.

Derivatives of Exponential Functions

The derivative of an exponential function reveals the rate at which the function's value changes with respect to a change in the input value. For exponential functions, the derivative retains the same form as the function itself, which is a unique characteristic not seen in other types of functions.

For the natural exponential function $f(x) = e^x$, the derivative is $f'(x) = e^x$. This means the rate of change of the function at any point is equal to the value of the function at that point.

For a general exponential function $g(x) = a^x$, the derivative involves the natural logarithm of the base a: $9^1(x) = a^x \, log_n \, a$ .This highlights the intrinsic relationship between exponential functions and logarithms.

These properties of the derivative of exponential functions are pivotal in solving differential equations and analysing change, making exponential functions indispensable in mathematical modelling and real-world applications.

Exponential functions, with their distinct properties and behaviours, are indispensable tools in mathematics and its applications. Their ability to model growth and decay processes, coupled with their unique derivative properties, makes them central to the fields of calculus, economics, and the natural sciences. Understanding the definition, properties, and derivatives of exponential functions equips one with the foundational knowledge required to tackle a broad spectrum of problems in both theoretical and applied mathematics.

8. SUMMARY

Wrapping up Unit 6, "Some Special Functions Part 1," we've embarked on a mathematical journey through the realms of Radian Measures and Trigonometric Functions. These aren't just abstract concepts; they're the language of the universe, describing everything from the simplest rotations to the complex oscillations in electrical engineering.

Starting with radian measures, we've seen how they offer a more natural way to describe angles, especially in mathematics and engineering. The idea that an angle's measure is directly related to the arc it subtends gives us a direct link between linear and angular measurements, making calculations more intuitive. It's like realizing that every time we talk about angles, we're talking about the journey around a circle. This concept isn't just academic; it's pivotal in fields like physics and computer graphics, where rotations and circular motions are bread and butter.

Moving on to trigonometric functions, we delved into the sine, cosine, and tangent functions, which are the heroes of trigonometry. These functions do more than just solve triangles; they describe the fundamental relationships between the sides and angles of a triangle, encapsulating the essence of circular motion and harmonic oscillations. For instance, when we look at the sine wave, we're not just seeing a mathematical curve; we're seeing the shape of sound waves, the rhythm of the tides, and even the pattern of daylight across the seasons.

The magic of these functions doesn't stop there. We learned about their reciprocal functions - secant, cosecant, and cotangent, and how they play their part in expanding our trigonometric toolkit. These functions might seem less intuitive at first, but they provide alternative perspectives on angles and ratios, enriching our understanding and problem-solving abilities.

But what truly ties this all together is the practical applications. Whether it's in engineering, computer science, or even economics, the ability to translate rotational movements into mathematical models or to understand the periodic nature of a phenomenon is invaluable. These concepts form the foundation for more complex ideas like Fourier analysis, which breaks down complex waves into simpler sine and cosine waves, a technique ubiquitous in signal processing and data analysis.

So as we wrap up this unit, let's not just see these as abstract mathematical concepts, but as the language through which we describe and understand the world around us. Whether you're analysing the stock market, designing a video game, or just trying to understand the physics of a spinning wheel, the knowledge we've gained here will be your guide.

9. SELF-ASSESSMENT QUESTIONS

SELF-ASSESSMENT QUESTIONS – 1

1. What is the measure of 1 radian in degrees?
   a) 57.3°
   b) 180°
   c) 90°
   d) 45°
2. If the radius of a circle is equal to the length of the arc, then the angle subtended by the arc at the centre is:
   a) 45°
   b) 1 radian
   c) 90°
   d) 180°
3. Which of the following is the correct expression for the cosine of an angle in the unit circle?
   a) $\cos(\theta) = y/r$
   b) $\cos(\theta) = x/y$
   c) $\cos(\theta) = x/r$
   d) $\cos(\theta) = r/y$
4. For the trigonometric function $\sin(\theta)$, what is the range of θ in the unit circle for its principal value?
   a) $[0, \pi/2]$
   b) $[-\pi/2, \pi/2]$
   c) $[0, \pi]$
   d) $[-\pi, \pi]$

5. If $\sin(\theta) = 1/2$, what is $\cos(\theta)$?
   a) $\sqrt{3}/2$
   b) $1/\sqrt{2}$
   c) $1/2$
   d) 0
6. What is $\tan(45°)$?
   a) 1
   b) $\sqrt{2}$
   c) 0
   d) $\sqrt{3}$
7. Which of the following is a reciprocal trigonometric function of sine?
   a) Cosine
   b) Secant
   c) Cosecant
   d) Tangent
8. What is the derivative of the function $f(x) = e^x$?
   a) e^x
   b) xe^{x-1}
   c) $e^x \log_n (x)$
   d) $x\, e^x$
9. If $f(x) = \log_n (x)$, what is the derivative $f'(x)$?
   a) $1/x$
   b) $\log_n(x)$
   c) $x \log_n(x)$
   d) $1/x^2$

10. For the function $f(x) = \sin(x)$, what is the derivative $f'(x)$?
    a) $\cos(x)$
    b) $\sin(x)$
    c) $-\sin(x)$
    d) $-\cos(x)$
11. Calculate the radian measure of a 30° angle.
12. Express $3\pi/4$ radians in degrees.
13. If $\cos(\theta) = 0.5$, find the value of θ in radians.
14. Find the value of $\sin(\pi/6)$.
15. Calculate the value of $\cot(\pi/4)$.
16. Determine the derivative of $f(x) = 3x^2$.
17. If $y = \log(x)$, find the derivative dy/dx.
18. Calculate the derivative of $f(x) = e^{(2x)}$.
19. Find the value of $\sec(60°)$.
20. If $\tan(\theta) = 1$, find the value of θ in degrees.

INSPIRED BY LIFE

10. TERMINAL QUESTIONS

1. Convert 120° into radians.
2. Find the radian measure of an angle that subtends an arc of length 8 units on a circle with a radius of 2 units.
3. Express $\pi/3$ radians in degrees.
4. If $\sin(\theta) = 0.866$, find the principal value of θ in degrees.
5. Determine the value of $\cos(\pi/4)$.
6. Calculate the value of $\tan(\pi/6)$.
7. Find the cosecant of an angle whose sine is 1/2.
8. Determine the secant of an angle whose cosine is $\sqrt{3}/2$.
9. If $\cot(\theta) = \sqrt{3}$, find the principal value of θ in radians.
10. Calculate the derivative of the function $f(x) = x^3 - 6x^2 + 11x - 3$.
11. Find the derivative of the function $f(x) = \log_n (x^2)$.
12. Determine the derivative of the function $f(x) = e^{(3x)}$.
13. Calculate the derivative of $f(x) = \sin(2x)$.
14. Find the value of $\sin^{-1}(1/\sqrt{2})$.
15. Calculate the value of $\tan^{-1}(1)$.
16. Determine the derivative of the function $f(x) = \log(x)$ (base 10).
17. Find the derivative of the function $f(x) = 3e^{(-2x)}$.
18. Calculate the value of $\sec^{-1}(2)$.
19. If $y = x \sin(x)$, find dy/dx using the product rule.
20. Determine the derivative of the function $f(x) = \cos(x^2)$.

11. ANSWERS

Self-Assessment Questions

1. A) 57.3°
2. B) 1 radian
3. C) $\cos(\theta) = x/r$
4. B) $[-\pi/2, \pi/2]$
5. A) $\sqrt{3}/2$
6. A) 1
7. C) Cosecant
8. A) e^x
9. A) $1/x$
10. A) $\cos(x)$
11. $\pi/6$ radians
12. 135°
13. $\pi/3$ radians
14. 1/2
15. 1
16. 6x
17. 1/x
18. 2e^(2x)
19. 2
20. 45°

Terminal Questions

1. $2\pi/3$ radians
2. 4 radians
3. 60°
4. 60° (θ could be in the first or second quadrant, but the principal value is 60°)
5. $\sqrt{2}/2$
6. $\sqrt{3}/3$
7. 2
8. $2/\sqrt{3}$

9. $\pi/6$
10. $3x^2 - 12x + 11$
11. $2/x$
12. $3e^{(3x)}$
13. $2\cos(2x)$
14. $\pi/4$ radians
15. $\pi/4$ radians
16. $1/(x \log_n (10))$
17. $-6e^{(-2x)}$
18. $\pi/3$ radians (assuming the principal value branch for sec^(-1))
19. $\sin(x) + x \cos(x)$
20. $-2x \sin(x^2)$

12. REFERENCES

1. Homer, M. (n.d.). *Trigonometry: Graphing the Sine, Cosine and Tangent Functions*. Owlcation. https://owlcation.com/stem/Trigonometry-graphs-of-sin-x-cos-x-and-tan-x
2. *Trigonometry -- Graph*. (n.d.). Sngweb.com. Retrieved March 26, 2024, from https://sngweb.com/education/formula/trigonometric/graph.html
3. (2015). Trigidentities.net. https://trigidentities.net/wp-content/uploads/2022/01/Inverse-of-Trigonometric-Functions.jpeg